

Customer Purchase Prediction

Using Logistic Regression and Random Forest
Machine Learning Algorithms

ML Presentation

Presented by :

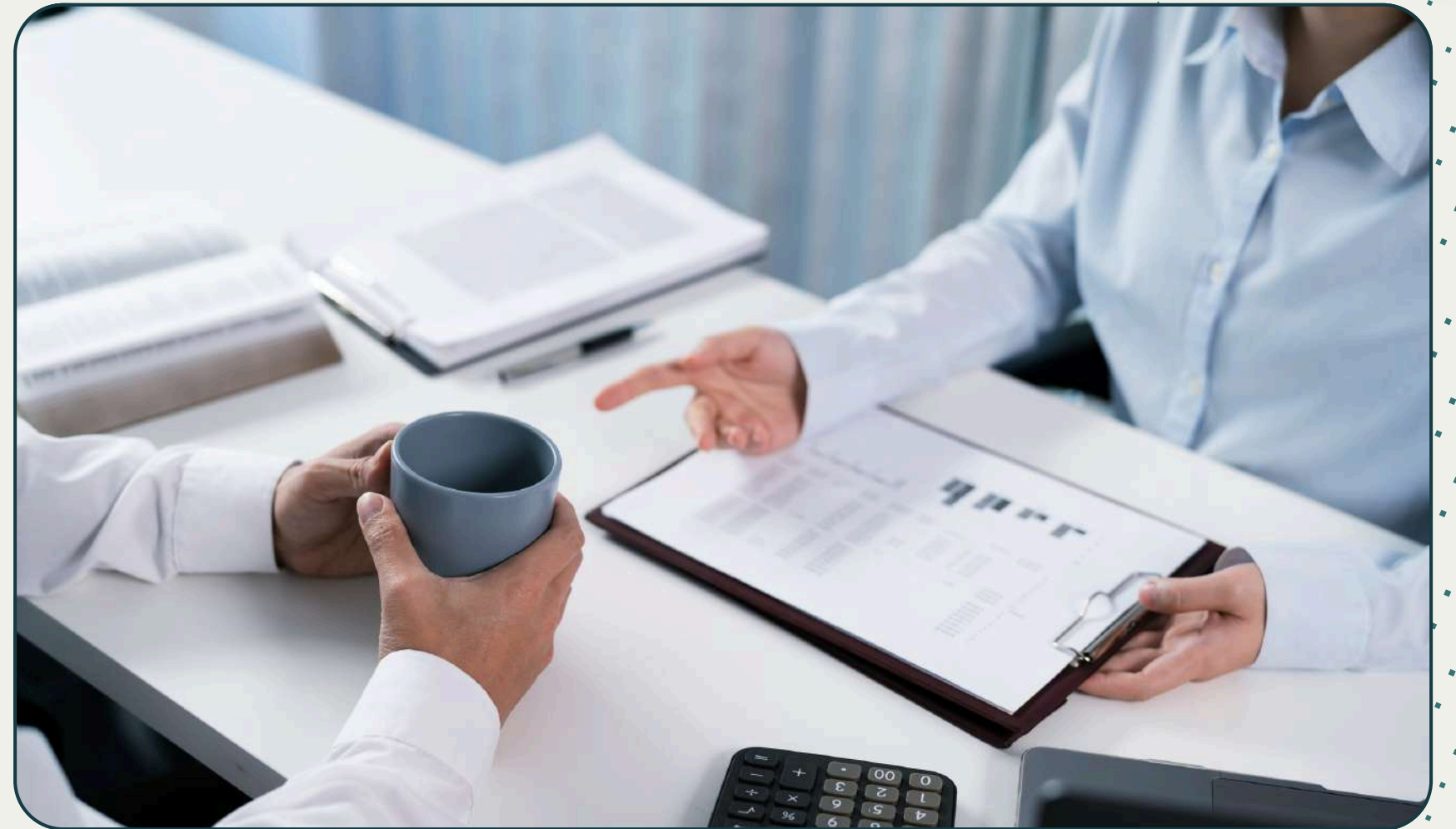
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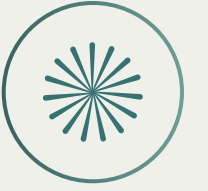
Abstract

Customer purchase prediction is an important application of machine learning that helps businesses identify whether a customer is likely to purchase a product based on demographic and behavioural information. This project develops and compares two supervised machine learning algorithms — Logistic Regression and Random Forest — for predicting customer purchase behaviour.

The dataset is preprocessed by handling missing values, encoding categorical variables, scaling features, and splitting the data into training and testing sets. The models are evaluated using accuracy, precision, recall, F1-score, and confusion matrix to compare their predictive performance.



Problem Statement



- Businesses collect large amounts of customer demographic and behavioural data.
- It is difficult to identify which customers are likely to purchase a product.
- Traditional methods may not effectively identify hidden patterns and relationships in customer data.
- An automated prediction system is needed to classify customers based on their purchase likelihood.
- Different machine learning algorithms may produce different levels of prediction performance.
- Therefore, this project aims to implement and compare Logistic Regression and Random Forest for customer purchase prediction.

Objectives & Dataset

Our goals and the data behind the prediction



Objectives

- 01 Predict customer purchase behaviour using machine learning.
- 02 Preprocess and prepare customer data for model training.
- 03 Implement Logistic Regression and Random Forest.
- 04 Evaluate both models using multiple performance metrics.
- 05 Compare the predictive performance of the two algorithms.
- 06 Explore how purchase prediction can support targeted marketing and customer engagement.



Dataset

Total Records

1,000

Number of Features

6

Target Variable

Purchase
(Yes / No)



Input

Demographic and behavioural customer data



Features

- Age
- Gender
- Income
- Browsing History
- Previous Purchases
- Time Spent on Website

DATA PREPROCESSING

01 — Data Cleaning

- Checked the dataset for missing values and duplicate records.
- Handled incomplete or inconsistent data.
- Removed unnecessary columns that do not contribute to prediction.

02 — Categorical Data Encoding

- Converted categorical variables into numerical values.
- Example:
 - Gender → Male/Female → 0/1
- This allows ML algorithms to process categorical information.

03 — Feature Selection

- Selected relevant demographic and behavioural attributes that can influence purchase decisions.
- Removed irrelevant or redundant features where required.

04 — Feature Scaling

- Numerical features were scaled to bring them into a comparable range.
- This is particularly useful for Logistic Regression, where differences in feature magnitude can affect the model.

05 — Train-Test Split

- The processed dataset was divided into: Training Set → used to train the models
- Testing Set → used to evaluate model performance
- A typical split is 80% training / 20% testing (use your actual split if different).

Data Preprocessing

Flow
Raw Customer Data → Data Cleaning → Feature Selection → Categorical Encoding → Feature Scaling → Train-Test Split → Preprocessed Dataset → Model Training

Data Preprocessing

Before training the models, the raw customer dataset is transformed into a clean and suitable format for machine learning.

System Architecture

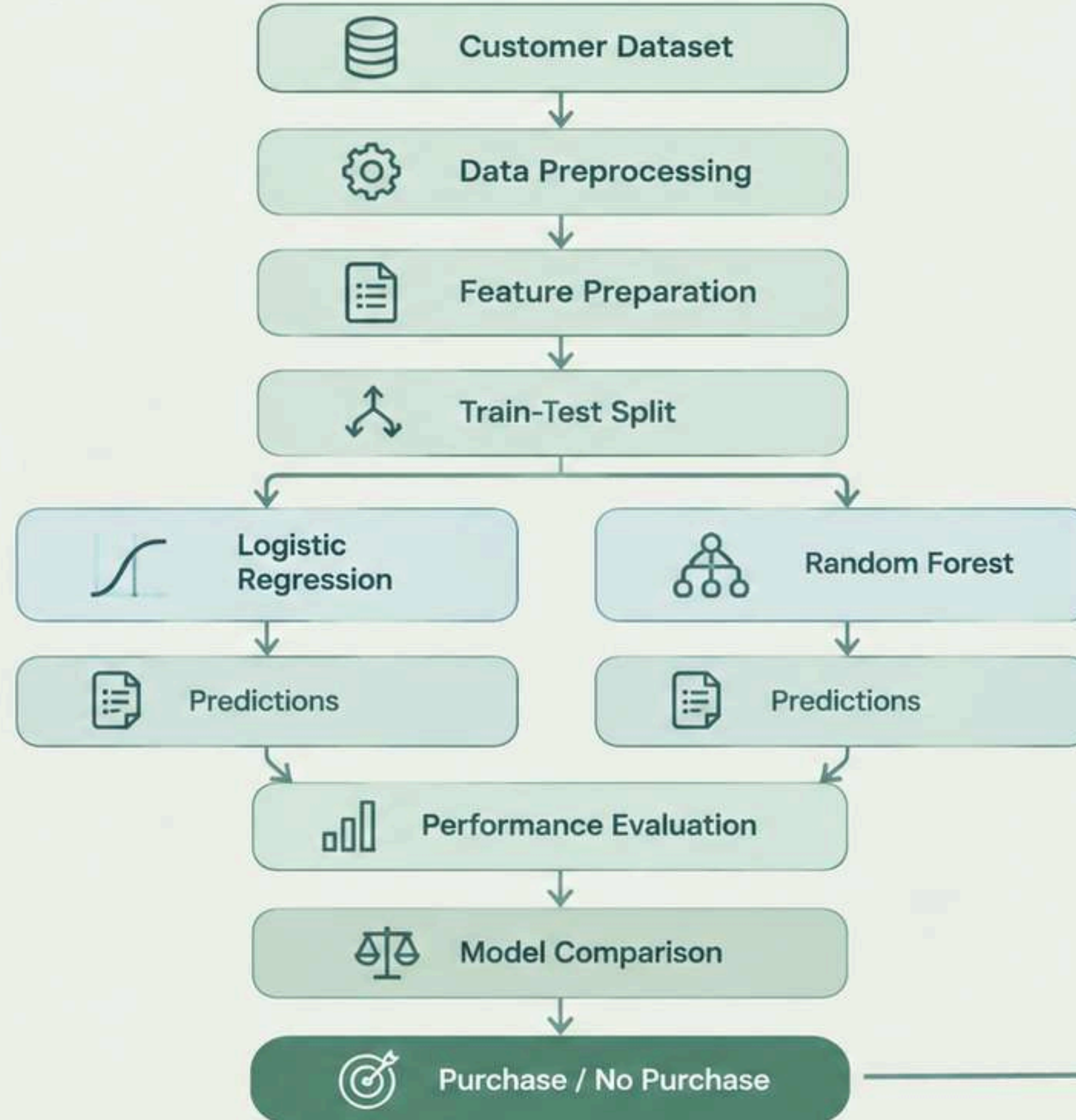
From customer data to purchase prediction

System Input



Customer demographic
+ behavioural
attributes

- Age
- Gender
- Income
- Browsing History
- Previous Purchases
- Time Spent on Website



System Output



Predicted customer
purchase behaviour



Machine Learning Implementation

Building and training the models for customer purchase prediction



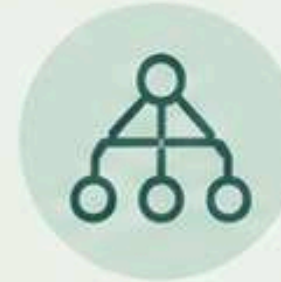
Logistic Regression

A supervised classification algorithm that estimates the probability of a customer belonging to the purchase or no-purchase class.



Implementation

- 01 Load preprocessed data.
- 02 Train the Logistic Regression model.
- 03 Generate predictions on test data.
- 04 Calculate performance metrics.



Random Forest

An ensemble classification algorithm that combines multiple decision trees to generate a final prediction.



Implementation

- 01 Train multiple decision trees using the training data.
- 02 Generate predictions from individual trees.
- 03 Combine predictions using majority voting.
- 04 Evaluate the final predictions.



Technologies Used



Python



Pandas



NumPy



Scikit-learn



Matplotlib / Seaborn

Model Evaluation & Results

Assessing performance and comparing the two models



Performance Evaluation

The models are evaluated using the following metrics:



Accuracy

Measures the overall percentage of correct predictions.



Precision

Measures how many predicted purchasers were actually purchasers.



Recall

Measures how many actual purchasers were correctly identified.



F1-Score

Provides a balance between precision and recall.



Confusion Matrix

Shows correct and incorrect predictions for each class.

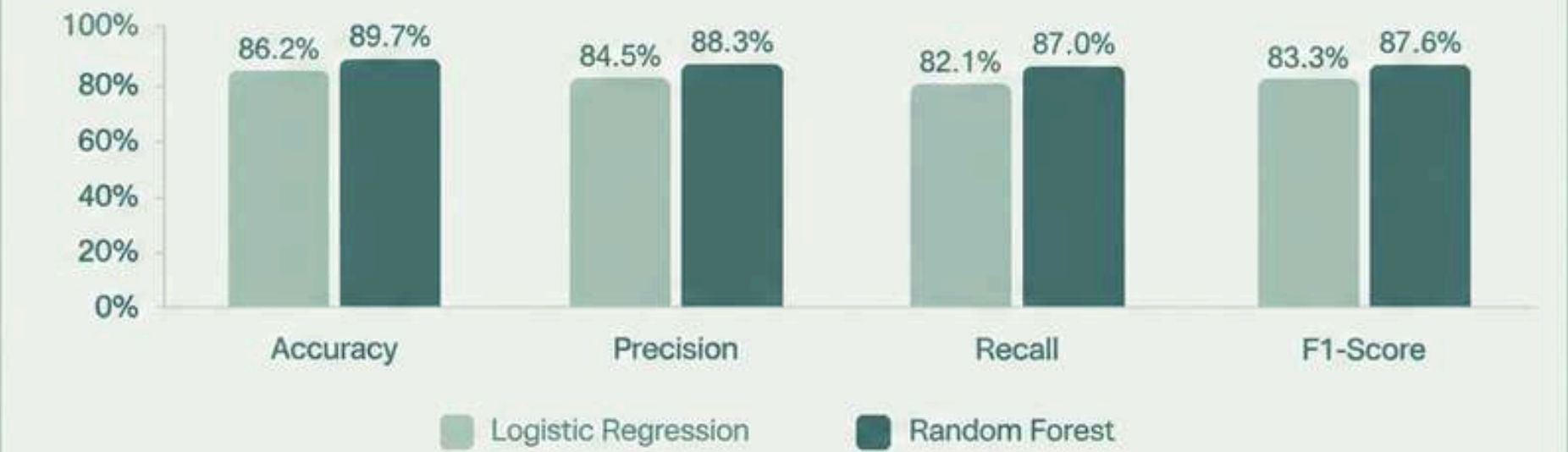


Results

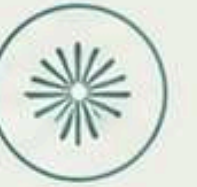
Metric	Logistic Regression	Random Forest
Accuracy	86.2%	89.7%
Precision	84.5%	88.3%
Recall	82.1%	87.0%
F1-Score	83.3%	87.6%



Performance Comparison



Applications & Future Scope



Applications



Targeted Marketing

Identify customers with higher purchase likelihood.



Personalized Promotions

Recommend relevant offers and products.



Customer Insights

Understand purchasing patterns and behaviour.



Business Decision Support

Support data-driven marketing strategies.



Future Scope



Larger & Diverse Data

Improve generalization with richer datasets.



Feature Expansion

Include browsing, engagement and transaction signals.



Model Optimization

Apply hyperparameter tuning and advanced ML models.



Real-Time Deployment

Enable instant purchase prediction through a web application.

Conclusion



- A machine learning-based system was developed to predict customer purchase behaviour.
- Customer data was preprocessed through cleaning, encoding, scaling and train-test splitting.
- Logistic Regression and Random Forest were implemented for classification.
- Both models were evaluated using accuracy, precision, recall, F1-score and confusion matrix.
- The performance comparison provides insights into the suitability of different ML approaches for purchase prediction.
- The system demonstrates how machine learning can support targeted marketing and data-driven business decisions.



FINAL TAKEAWAY

Customer Data → ML Models → Purchase Prediction → Business Insights



Thank
You